RIGHT MULTI-FUNCTION SWITCH (Continued)

OPERATION

The right multi-function switch uses an internal logic circuit, an internal relay and conventionally switched outputs to control the many functions and features it provides. The switch receives battery current on a fused ignition switch output (run-acc) circuit from a fuse in the Power Distribution Center (PDC) and the wiper and washer system fuse in the fuse block whenever the ignition switch is in the On or Accessory positions. The electronic intermittent wipe logic circuit within the switch receives a logic ground at all times through the body wire harness from a take out with an eyelet terminal connector secured under a ground screw on the left inner cowl side panel, below the left end of the instrument panel.

Following are descriptions of how the right multi-function switch control stalk and control knob are operated to control the functions and features they provide. Refer to the owner's manual in the vehicle glove box for more information on the features, use and operation of the right multi-function switch.

- **Front Wash Mode** - Pulling the control stalk of the right multi-function switch towards the steering wheel actuates the momentary single pole, single throw front washer switch and operates the front washer pump/motor.
- **Front Wipe Modes** - Moving the control stalk of the right multi-function switch up or down to one of four detent positions actuates the triple pole, quadruple throw front wiper switch and selects the Off, Low, High, or Delay front wiper operating mode. Moving the control stalk downward from the Off position actuates a momentary single pole, single throw switch that selects the Mist operating mode.
- **Intermittent Front Wipe Delay Intervals** - Rotating the control knob on the end of the control stalk actuates the front wiper delay variable resistor and selects the wiper delay interval when the Delay mode is selected with the front wiper switch control stalk.

The intermittent wipe logic circuit within the right multi-function switch monitors inputs from the front wiper switch circuitry, the front washer switch circuitry, the front wiper delay variable resistor, and the front wiper motor park switch. The programming of the logic circuit then determines the proper outputs to the front wiper motor. The low current logic circuit controls the high current front wiper motor by pulling the control coil of the integral wiper motor relay to ground. The wiper motor relay switches a circuit that feeds battery current to the low speed or high speed brush of the front wiper motor through the appropriate right multi-function switch circuits.

DIAGNOSIS AND TESTING - RIGHT MULTI-FUNCTION SWITCH

Be certain to perform the diagnosis for the front wiper system and/or front washer system before testing the right multi-function switch. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - DIAGNOSIS AND TESTING). Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the right multi-function switch from the steering column and disconnect the body wire harness connector for the switch from the switch connector receptacle.

(3) Using an ohmmeter, check the right multi-function switch continuity at the terminals (Fig. 20) in the switch connector receptacle as shown in the Right Multi-Function Switch Tests table.

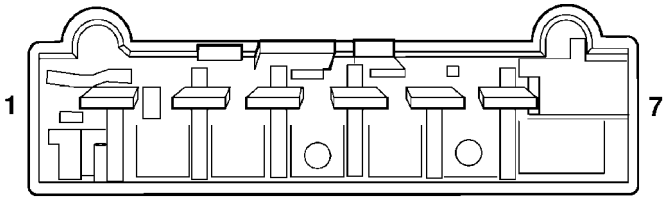

Fig. 20 Right Multi-Function Switch Connector Receptacle